

Human Robot Interaction for Blind and Low Vision People: A Systematic Literature Review

Yize Wei*
Computer Science
National University of Singapore
Singapore
yize.wei@u.nus.edu

Mia Huong Nguyen
Augmented Human Lab
National University of Singapore
Singapore
mhnguyen@u.nus.edu

Nathan Rocher
IRIT
University of Toulouse 3
France
nathan.rocher@irit.fr

Roger Zimmermann
Computer Science
National University of Singapore
Singapore
rogerz@comp.nus.edu.sg

Chitralekha Gupta
Augmented Human Lab
National University of Singapore
Singapore
chitralekha@nus.edu.sg

Wei Tsang Ooi
Computer Science
National University of Singapore
Singapore
ooiwt@comp.nus.edu.sg

Christophe Jouffrais
IPAL, CNRS
Singapore
christophe.jouffrais@cnrs.fr

Suranga Nanayakkara
Augmented Human Lab
National University of Singapore
Singapore
suranga@ahlab.org

ABSTRACT

Recent years have witnessed a growing interest in using robots to support Blind and Low Vision (BLV) people in various tasks and contexts. However, the Human-Computer Interaction (HCI) community still lacks a shared understanding of what, where, and how robots can benefit BLV users in their daily lives. In light of this, we conducted a systematic literature review to help researchers navigate the current landscape of this field through an HCI lens. We followed a systematic multi-stage approach and carefully selected a corpus of 76 papers from premier HCI venues. Our review provides a comprehensive overview of application areas, embodiments, and interaction techniques of the developed robotic systems. Further, we identified opportunities, challenges, and key considerations in this emerging field. Through this systematic review, we aim to inspire researchers, developers, designers, and HCI practitioners, to create a more inclusive environment for the BLV community.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Human-robot Interaction, People with Visual Impairments, Assistive Technology, Systematic review

* Authors Yize Wei, Nathan Rocher, and Wei Tsang Ooi are also with IPAL, CNRS. Author Christophe Jouffrais is also with IRIT, CNRS.

ACM Reference Format:

Yize Wei, Nathan Rocher, Chitralekha Gupta, Mia Huong Nguyen, Roger Zimmermann, Wei Tsang Ooi, Christophe Jouffrais, and Suranga Nanayakkara. 2025. Human Robot Interaction for Blind and Low Vision People: A Systematic Literature Review. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26-May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 19 pages. <https://doi.org/10.1145/3706598.3713438>

1 INTRODUCTION

Globally, visual impairments affect at least 2.2 billion people¹. To improve the quality of life for the Blind and Low Vision (BLV) community, the field of Human-Computer Interaction (HCI) has seen a myriad of assistive technologies designed and created². Historically, these technologies for BLV users have predominantly taken the form of wearable devices and smartphone applications.

With the development of robotics and connected fields, robots have transitioned from once futuristic concepts to common presence in daily life. For example, service robots in restaurants now greeting patrons and delivering food to the table. This evolution has paved the way for assistive robots, which have now become an active research field. For BLV users, robotic canes have been inspired by traditional white canes [19, 122], and robotic guide dogs have been developed as a model of traditional guide dogs [31, 55]. On one hand, these endeavors demonstrate the immense potential of robotic technologies to enhance the independence and overall well-being of BLV users, evidenced by the growing number of papers in this field in recent years (refer to Figure 5); on the other hand, they also reveal design tensions in creating user-friendly

¹<https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment>

²We understand it is important to use neutral and appropriate language to describe people with disabilities, as it promotes respect and inclusivity. Following the recommendations from the SIGACCESS community [50], we use the term “*Blind and Low Vision (BLV) people*”. We also acknowledge that other terms, such as “*People with Visual Impairments (PVI)*”, are also appropriate.

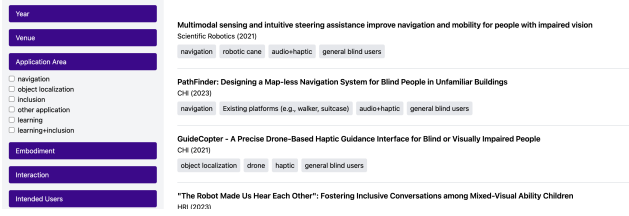


Figure 1: Screenshot of the online paper library showcasing all the papers in our corpus. The website can be accessed at <https://slr-paper.vercel.app/slr-paper>.

assistive robots, due to the complexity of robotic systems and the unique needs of this demographic.

Despite growing interest, there is a lack of shared understanding of what, where, and how robots can be used to assist BLV users in their daily lives, which limits our ability to fully harness the potential of robotic technology in supporting this demographic. We believe bridging this gap is both timely and crucial, as a systematic review could help researchers better navigate the current landscape by providing insights into the opportunities, challenges, and best practices for designing assistive robots for BLV users. Specifically, our review aims to address these overarching research questions:

- **RQ1:** What are the potential research opportunities in human-robot interaction for BLV users?
- **RQ2:** What are the key considerations in designing assistive robots for BLV users?
- **RQ3:** What are the challenges in conducting research in this field, and how should we approach them?

To answer these questions, following the methodology proposed by Kitchenham [67], we conducted a systematic literature review of robots designed and developed for BLV users. We identified a final corpus of 76 papers published in leading HCI and robotic venues from 1981 to 2024. We provided a detailed analysis of the robotic systems, with a particular focus on the application areas, embodiments, input mechanisms, and output mechanisms of the robotic system. We also reported on methodological decisions that have been made by other researchers along the journey. We ended with a discussion on the challenges and opportunities for future research. Our findings can serve as a valuable resource for researchers, designers, and HCI practitioners working on assistive robots to support BLV users.

In summary, this paper makes the following contributions: (1) a typology of assistive robots designed for BLV users and a discussion on current trends, including application areas, embodiments, and interaction mechanisms of these robots; (2) identification of gaps and opportunities for future research; (3) and an online paper library containing 76 papers that we identified in this review (as shown in Figure 1).

2 BACKGROUND

2.1 Overview of Review Papers on Assistive Robots

Assistive robots have been the focus of HCI researchers and practitioners for many years. In this section, we step back and provide an

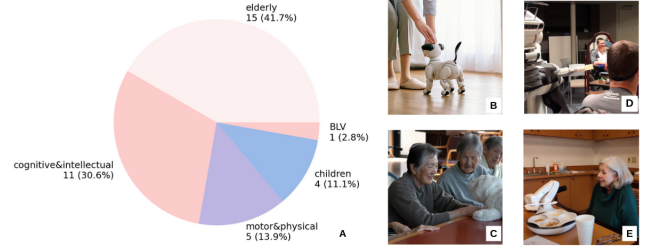


Figure 2: A) The distribution of review papers on assistive robots. Our categorization is aligned with previous survey papers in the HCI literature [85, 98]. B) AIBO. C) PARO. D) Robot-assisted scratching. E) Robot-assisted feeding. Courtesy to the authors.

overview of review papers that focus on assistive robots in general to situate our work in the broader literature.

In total, we found 36 relevant review papers on assistive robots that are tailored to children, the elderly, and those with disabilities (Figure 2 (A))³. Among these review papers, 15 (42%) focused on the use of assistive robots for elderly care, 17 (47%) focused on their use for people with disabilities, and 4 (11%) focused on their use for children. Specifically, only one (2.8%) review paper focused on BLV users.

As we mentioned just above, a large portion of papers focused on elderly care. Some notable commercial robots include AIBO⁴ and Paro⁵, which resemble dogs and seals respectively (as shown in Figure 2 (B-E)). These robots are designed to provide elderly with physical assistance in accomplishing daily tasks, and psychological support [1]. The use of assistive robots has been explored in the realm of cognitive (e.g., dementia) and intellectual disabilities (e.g., autism, cerebral palsy) as well. These robots, similarly, place a strong emphasis on providing emotional and affective support [27].

In contrast, we found significantly fewer review papers on assistive robots for people with motor impairments and BLV users. In both cases, these robots present unique opportunities as they empower users to access the physical world. For example, these robots can serve as surrogates for body parts for people with motor impairments in performing daily tasks such as scratching [30] or eating [15]. Of particular relevance to our work is a brief review by Kyrarini et al. [71] that focused specifically on robotic assistive technologies for BLV users. However, their final corpus was limited to papers published between 2021 and 2023 due to the method they used. To the best of our knowledge, there is no consolidated analysis on what (e.g., embodiment), where (e.g., application area), and how (e.g., interaction modalities) robots can be used to assist BLV users in their daily lives. We provide a systematic review of papers (refer to Section 3) from a wider time frame, aiming to offer a more extensive and inclusive perspective on this topic.

³We have used the search string TITLE (("review" OR "survey") AND "assistive robot*") in the Scopus database. We excluded papers that do not present a review of assistive robots focused on a specific demographic (e.g., papers that present results from online surveys) or were not written in English.

⁴<https://us.aibo.com/>

⁵<http://www.parorobots.com/>

The use of assistive robots in human society has also stirred up discussions on various ethical and societal concerns in the literature. Currently, privacy issues, trust, and autonomy are the major challenges [16, 116]. For example, an excessive amount of assistance is inappropriate and even harmful, as it undermines the BLV users' ability to make decisions. At this junction, it is pivotal for HCI researchers and practitioners to know how to design interfaces that foster healthy and balanced relationships between the BLV users and assistive robots; and, to take it one step further, consider the relationship between assistive robots and other stakeholders (e.g., bystanders). We have studied and consolidated our insights in this systematic literature review.

2.2 Assistive Robots for Blind and Low Vision Users

BLV people face different challenges in their lives. Main challenges they face is navigation. While Orientation & Mobility training and mobility aids such as canes are helpful, BLV people often require additional navigation tools to assist with self-orientation and wayfinding. In their daily lives, they also struggle with other tasks such as reading text and diagrams, as well as locating people and objects [21, 35, 47]. Importantly, these challenges not only manifest differently across various contexts (e.g., education, work, or leisure), but also vary significantly between person to person, due to factors such as vision abilities, prior experiences, and individual ability to access support systems.

For most BLV people, the primary mobility aids for BLV people are guide dogs and white canes, both of which have drawbacks. White canes can only detect obstacles or changes to the terrain a few steps ahead, and cannot address issues related to orientation. Guide dogs are scarce due to the extensive breeding, training, and matching process [81], and they cannot detect overhead obstacles. In recent years, the evolution of technology has given birth to a myriad of assistive technologies.

Robots offer distinct advantages that extend beyond the capabilities of wearable devices (e.g., rings [117], belts [100], gloves [88], smart headphones [17, 18]) and smartphone applications. Their larger form factor provides the space for more sensors and computational units, and hence more advanced capabilities [26]. Moreover, robots typically possess some degree of autonomy and mobility, which make them more versatile assistants for BLV users in various contexts (as opposed to wearable devices that are attached to the human body). Finally, robots are in this unique position where they are viewed as both “things” and “social agents” [4]. This raises interesting discussions about the relationship between assistive technologies and BLV users.

MELDOG was among the pioneering works in this field [127, 128]. Benefiting from technological maturity in both hardware and software, more robots have been designed and developed for a broader range of applications [40, 49, 56]. These robots look and function differently, and are intended to be used in diverse contexts and tasks. Furthermore, these studies are conducted by researchers from diverse backgrounds using heterogeneous research methods; their publications scattered across multiple venues such as CHI. While this field has developed significantly over the years, more research is needed for a shared understanding in the HCI community

in terms of what (e.g., embodiment), where (e.g., application area), and how (e.g., interaction modalities) robots can be used to assist BLV users in their daily lives. This shared understanding and knowledge are crucial for researchers and practitioners to collectively advance this field and build a more inclusive future.

3 METHODOLOGY

3.1 Overview

There are two primary literature review strategies, which are systematic and narrative [33]. A systematic literature review aims to provide a *comprehensive* and *unbiased* overview of a given topic, whereas a narrative review does not necessarily describe the methodological decisions in the searching process [33]. We chose the systematic approach as we would like to help researchers navigate the current landscape of this emerging field via a *reproducible* manner. We conducted the systematic literature review following the practices recommended by Kitchenham [67], as it is extensively used for surveys in the field of computer science and beyond [25, 38].

The flowchart of the multi-stage process is shown in Figure 3. We would like to highlight that this process is a collective effort by a diverse team of researchers with backgrounds in HCI, AI, robotics, and psychology, most of whom have experience working with BLV users. In the next subsections, we elaborate on the key stages.

3.2 Defining Terms

The term “robot” is widely used in the literature (refer to Table 1). We therefore sense that there is a need to have a clear understanding of what we mean by “robot”. In this review, we chose to embrace the definition proposed by Breazeal et al. [87] as it aligns with our HCI perspective. We view robots as “Physical machines that are designed to interact with people in human-centric terms and to operate in human environments alongside people.” Following this definition, these robots are expected to feature a certain level of intelligence and programmability. In this review, we excluded publications that do not satisfy this definition. For example, we excluded wearable assistive devices or virtual agents that exert no agency in the physical world. However, assistive devices like robotic canes were included as canes can guide BLV users in various navigation tasks in the physical world.

3.3 Database Search

3.3.1 Search String Execution. We used Scopus⁶ as our literature search database. Scopus is a multi-disciplinary and comprehensive database of peer-reviewed literature, abstracts, and citations that includes scientific journals, books, and conference proceedings from a diverse range of publishers (e.g., IEEE, ACM). Moreover, Scopus provides tools to filter and visualize the selected publications, due to which it is a popular database for systematic literature reviews [25, 95]. Our search string was carefully crafted to combine two main concepts: “robots” and “BLV users”. We conducted the initial database search in December 2023, and the search obtained 1411 results. Later, we conducted a follow-up search to incorporate recently published papers as of August 2024, which obtained 140 new results. In total, search string execution yielded 1551 entries,

⁶<https://www.scopus.com/>

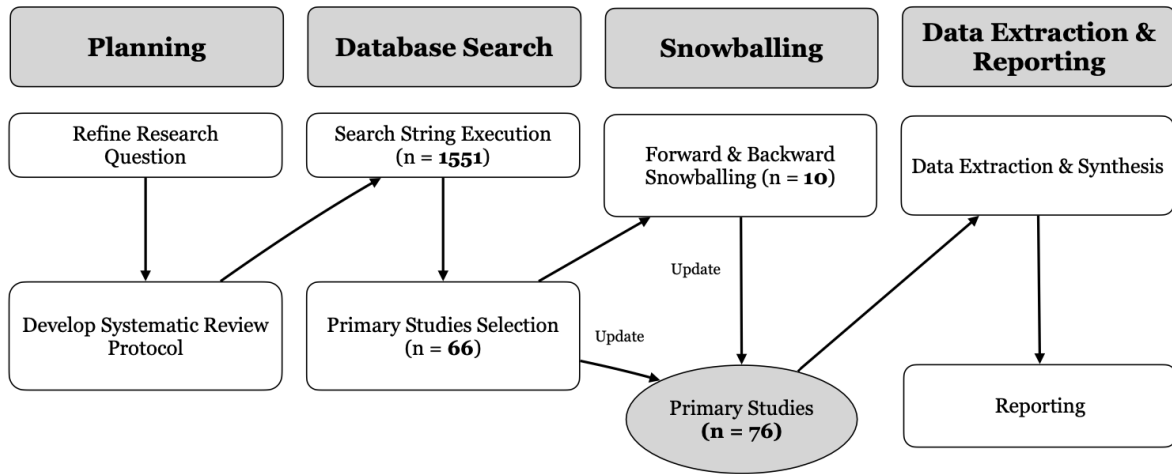


Figure 3: The multi-stage systematic literature review process.

Table 1: A list of other commonly recognized definitions for related terms.

Term	Definition & Source
<i>Robot</i>	- Robots are constructed systems that display both physical and mental agency but are not alive in the biological sense [111]. - A machine that resembles a living creature in being capable of moving independently (as by walking or rolling on wheels) and performing complex actions (such as grasping and moving objects). Oftentimes, such a machine is built to resemble a human being or animal in appearance and behavior (Webster's Dictionary).
<i>Assistive Robots</i>	Robots that give aid or support to a human user, especially for people with disabilities [41].
<i>Social Assistive Robots</i>	Robots that are capable of assisting users through social rather than physical interaction, just like a good coach or teacher [121].
<i>Socially Intelligent Robots</i>	Artificial agents which show elements of human-style social interaction and behavior [36].

TITLE-ABS-KEY
 (("robot*" OR "drone*" OR "unmanned aerial vehicle*" OR "UAV")
 AND
 ("the blind" OR "blind elderly" OR "blind children" OR "blind people" OR "blind person" OR "blind
 user*" OR "visual disabilit*" OR "visual* impair*" OR "visual*challenged" OR "low vision"))

To search in the title, abstract, and key words
 To prevent false positive results (e.g., medical publications related to blind studies), we refined the latter part of the search to specifically target the blind population

Figure 4: Search string with annotations.

which will then be further selected in the screening step. Next, we describe the Inclusion & Exclusion Criteria (Section 3.4) and detailed screening process (Section 3.5).

3.4 Inclusion & Exclusion Criteria

The team collectively established a set of Inclusion Criteria (IC) and Exclusion Criteria (EC) to select the papers further. The criteria are shown in Table 2. For EC1, EC2, EC3, and EC4, we filtered by *Language*, *Document Type*, and *Length* in the Scopus database respectively. For EC5, we first defined the scope of “*top venues*”, which is a common step in survey papers [25]. Given the interdisciplinary nature of the topic, we prioritized inclusivity in this review. To achieve this, we consulted various bibliometric sources, such as Clarivate⁷, Google Scholar⁸, and CORE ranking⁹ to make well-informed decisions. We chose venues that satisfied at least one of these criteria: h-index ≥ 40 , ranked as Top 20 publication venues in the field of HCI or Robotics according to Google Scholar, Impact Factor ≥ 3 , or CORE ranking of A or higher.

We also manually screened (see Section 3.5) the title and abstract of the papers to remove those who fell outside the scope of our review (which is related to EC6 and IC1). We excluded papers that develop other forms of assistive technologies. For example, we excluded wearable devices such as rings [117], smartglasses [42], or belts [101]. Similarly, we excluded papers that are primarily focused on voice interfaces [92] or mobile applications [29]. We also excluded the papers that propose only algorithms (e.g., obstacle avoidance algorithm [83]) that could potentially be incorporated into future robots, as these would divert our focus from the aforementioned HCI perspective.

Table 2: Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
IC1: Paper that designs, develops, or evaluates robots for blind users.	- EC1: Paper that is not in English. - EC2: Paper that is a tutorial, poster, or secondary study (e.g., survey). - EC3: Paper without an abstract or author details. - EC4: Paper that is less than four pages. - EC5: Paper that is not from a top venue. - EC6: Paper that, based on the title and abstract, is outside the scope of the review.

⁷<https://clarivate.com/>

⁸<https://scholar.google.com/>

⁹<https://portal.core.edu.au/conf-ranks/>

3.5 Screening

As mentioned earlier, we have screened the papers manually to select the papers that are aligned with our focus. To achieve this, three coders jointly evaluated the papers by screening the titles and abstracts of the papers. To ensure consistency, we first computed Interrater Reliability (IRR) using *Krippendorff's alpha* [51, 89], which is a statistical measurement designed to measure agreement between any number of coders when coding qualitative data. We chose a subset of papers ($N = 50$) and independently marked them as either *Accept*, *Reject*, or *Borderline*. The result is a 50×3 nominal matrix. The calculated Krippendorff's alpha for this matrix was 0.9, indicating a high level of agreement among the coders. Three coders independently coded the rest of the papers. The borderline cases were discussed and decided among the coders until a consensus was reached. At this stage, we arrived at 66 selected papers for review.

3.6 Snowballing

Next, we expanded our initial selected set of papers by forward and backward snowballing. First, we extracted the *referenced papers* (i.e., papers that have been cited in one of the previously selected papers) and the *citing papers* (i.e., papers that cite one of the previously identified papers). Then, we applied our Exclusion Criteria (EC) and Inclusion Criteria (IC) (refer to Section 3.4) again to these papers to select the pertinent ones. During the snowballing step, an additional 10 papers were identified. In summary, our review curated a final corpus of 76 papers published in leading HCI and robotic venues on this topic.

3.7 Data Extraction

To facilitate efficient data extraction, we created a spreadsheet. Each column title was clearly defined to ensure a shared understanding of the type of information we needed to extract. Please refer to Appendix A for the full data extraction table structure. Four co-authors independently extracted information from each paper and populated the spreadsheet. Similar to before, any borderline or uncertain cases were resolved through group discussion among the coders until consensus was achieved. After the initial data extraction, we performed a consistency check to identify missing entries and compare the extracted data across co-authors. The extracted data was used for further analysis and synthesis, which will be presented in the following sections.

4 FINDINGS

4.1 Overview

Our analysis begins with metadata, including publication trends, venues, and target user demographics. Next, we dive deeper into the assistive robots designed and developed in these papers, structuring along four dimensions inspired by prior surveys [87, 118]: application areas, embodiments, input, and output. We chose to structure our discussion in these dimensions to form a holistic picture of when, what, and how assistive robots can assist BLV users: application areas describe “*when*” these robots can be used; embodiments examine the physical affordances of the assistive robots, helping readers to understand “*what*” robots have been proposed;

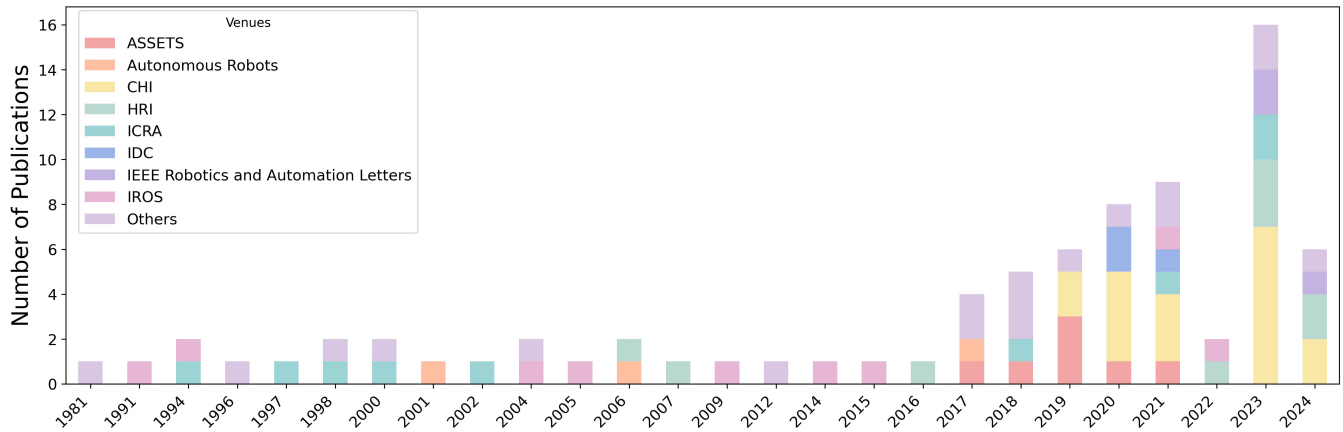


Figure 5: Number of publications in various venues over the years. The venues with fewer publications (≤ 2) are grouped under 'Others'. Note that the intervals between the years are not consistent due to space limitations (e.g., there are no publications between 1981 and 1991). Also, we included papers published up to August 2024, and more papers are expected to be published throughout the remainder of the year.

and interaction techniques shed light on *"how"* BLV users engage with the proposed systems. We also explore chronological trends and patterns to help future researchers and practitioners trace the field's intellectual lineage. Lastly, we report on the design and evaluation methods of these papers, which guided our discussion on the methodological challenges in this field.

4.2 Metadata

4.2.1 Publication Year. As shown in Figure 5, there is a substantial growth in the number of publications on this topic. This highlights the growing interest in exploring this area in recent years, and it also highlights the timeliness and relevance of this review, as it aims to provide a comprehensive overview to help researchers navigate the current landscape of this multidisciplinary field.

Interestingly, we observe an upsurge starting in 2017. One potential explanation is the impact of the significant developments in deep learning in that period. For example, the original paper on YOLO (You Only Look Once), a popular object detection algorithm, was published in 2016 [110]. This is evidenced by the considerable amount of studies that use deep learning algorithms as part of the implementation [32, 49, 84].

We also observe a drop in 2022, and we believe it is related to the global pandemic's impact on research - as the researchers, especially HCI researchers, faced significant challenges in interacting with participants (and each other) during the pandemic. In the realm of accessibility research, the involvement of end-users is often crucial to understanding their needs and evaluating assistive technologies (which we will elaborate more in Section 4.7.3). However, we can observe that the number of publications has rebounded significantly in 2023. This timeline aligns with the progression of the pandemic.

4.2.2 Publication Venue. The most popular publication venues for research on this topic are the International Conference on Human

Factors in Computing Systems (CHI) (23.7%), the International Conference on Robotics and Automation (ICRA) (11.8%), the International Conference on Intelligent Robots and Systems (IROS) (11.8%), the International Conference on Human-Robot Interaction (HRI) (11.8%), and International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)¹⁰ (9.2%). Noticeably, although we found the most papers published in the CHI conference among all venues, we only identified relevant CHI papers starting from 2019. This observation mirrors previous findings that accessibility research has grown substantially over the past few years in CHI conference [85]. Additionally, this distribution highlights the interdisciplinary nature of the field: CHI, ASSETS, and HRI are premier venues for HCI research, and ASSETS focuses specifically on accessibility studies; while ICRA and IROS are leading conferences in the field of robotics. It also indicates that the development in this field requires interdisciplinary collaborations across the traditional boundaries of established disciplines.

4.2.3 Target User Group. Among all the papers we examined, most of them focus on building robots for BLV users in general (54 papers, 78%), with only a few papers tailored to the needs of specific demographics. We found twelve (16%) papers that explored the use of robots for BLV children, oftentimes accompanied by their sighted peers, educators, or parents. We also found five (7%) papers that explored the use of robots for BLV elderly. The experiences of BLV users are shaped by various factors, including visual abilities (which can vary drastically), age, and age of onset. Hence, while general-purpose tools for BLV users can be helpful, we believe it is crucial that assistive technologies are tailored to specific user needs to be truly usable (refer to Section 5 for more discussion).

¹⁰ASSETS was initially called the "ACM Conference on Assistive Technologies".

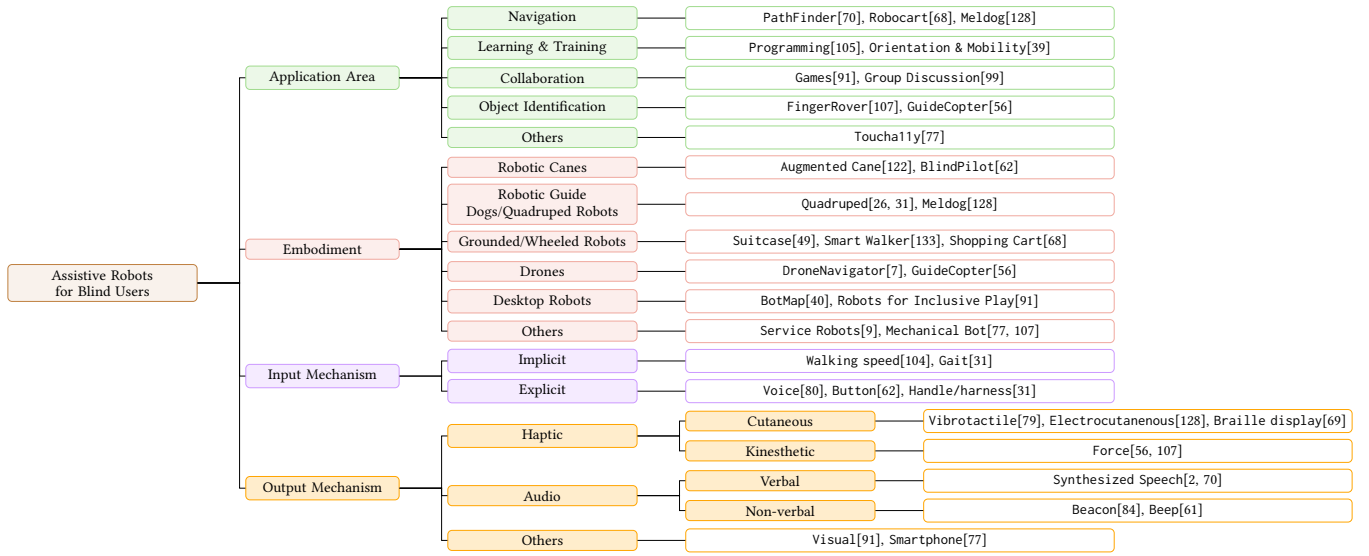


Figure 6: The typology of robots designed for BLV users in this survey with examples.

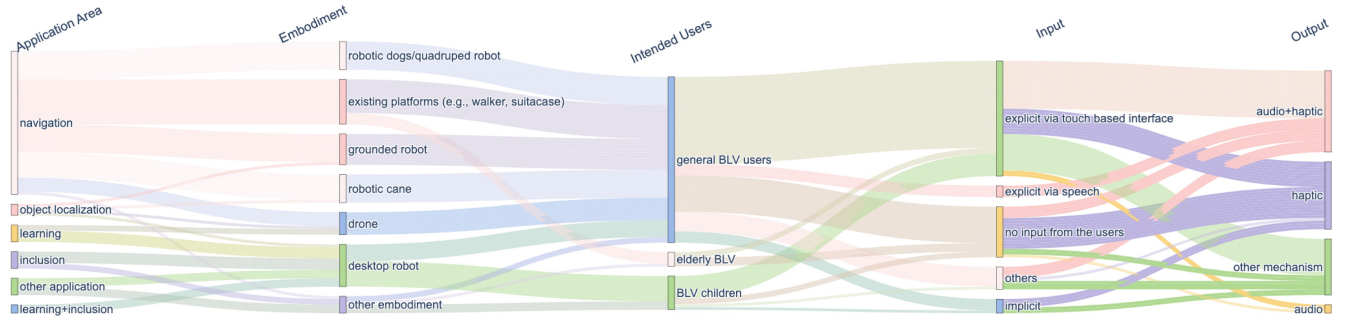


Figure 7: Inspired by other review papers [126, 129], we draw an Alluvial diagram for visualizing the flow of research papers from different application areas to various embodiments, intended users, and interaction mechanisms. Each node represents a unique category, and each link between nodes represents a single research paper, showing its journey across these categories. The diagram reveals that most of the robots created are focused on navigation for general BLV users and accept limited input from the users, highlighting a significant untapped potential for more interactive and personalized robots for BLV users.

4.3 Application Areas

Five main application themes emerged from the surveyed body of papers, which are Navigation, Learning & Training, Object Localization, Inclusion, and Others. In this section, we draw examples from the reviewed papers to provide insight into each category.

4.3.1 Navigation. Navigation emerges as the most extensively explored use case, with 51 (67%) papers addressing this theme. The identified papers studied navigation across various environments, including indoor spaces (e.g., museums [64] and grocery stores [68]), as well as different outdoor spaces [26, 79]. For example, robots have been used to guide BLV visitors to explore science museums more independently [64]. With the help of the robot, BLV visitors can explore the desired exhibits independently based on their interests and pace.

Navigation is one of the most challenging tasks for BLV people. It involves two main components. The first is *micro-navigation* or *mobility*, which means to sense the near-field environments via different senses (e.g., identify obstacles in the vicinity); the second is *macro-navigation* or *orientation*, which includes wayfinding, as the BLV users also need to concurrently process far-field environment to reach the destination [58, 82]. While robots show promise in assisting in both aspects of navigation, researchers need to understand the specific requirements of BLV users when developing these technologies.

4.3.2 Learning & Training. We found 9 (12%) papers related to learning and training. These two concepts are interrelated, as they both involve the acquisition and improvement of knowledge and skills. Recently, there has been a stream of works focused on using

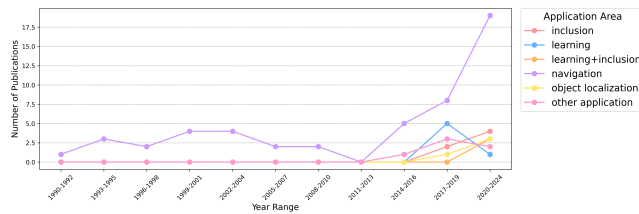


Figure 8: Historical Trend on Application Areas

robots, oftentimes desktop robots, as educational tools in classroom settings to help children with visual impairments learn abstract concepts such as programming or computational thinking [105, 113]. The benefits of using robots in these contexts go beyond providing physicality in the learning environments, they also encourage children with visual impairments to participate and collaborate, either with one another or with sighted peers (which we will elaborate on in Section 4.3.4). Additionally, robots, such as drones, are also used to help BLV people in Orientation & Mobility training, which is an essential skill for BLV people to navigate their environment safely and independently [39, 46].

4.3.3 Object Localization. One of the challenges BLV users face daily is locating objects or items, and we found 4 papers (5%) relating to this theme. Object localization differs from navigation (refer to Section 4.3.1) as it focuses on finding specific objects rather than guiding users through physical spaces. One example is GuideCopter. GuideCopter is a tethered drone that can help BLV users with fine-point object grasping tasks via haptic feedback (by exerting multi-dimensional pulling or pushing forces at fingertips) [56]. In comparison with an audio-based approach, Guidecopter has proven to be more efficient and accurate in completing the desired grasping task.

4.3.4 Inclusion & Collaboration. We found 9 (12%) papers related to inclusion and collaboration. Oftentimes, these robots are used to create activities that can enhance inclusion and collaboration between students with mixed visual abilities in classroom settings. This differs from learning as the goal is not to acquire knowledge but to promote participation. For example, Metela et al. have explored the use of robots to create an inclusive game experience for children with mixed visual abilities in a mainstream school [91], and this experience has proved to be engaging for children with different visual abilities.

4.3.5 Others. We also identified some other novel use cases 6 (8%). One example is Toucha11y, a mechanical bot that can increase the accessibility of public kiosks with touchscreens [77]. Once attached to the touchscreens with suction cups, Toucha11y can recognize and retrieve information, which will be rendered to the BLV user’s smartphone afterward.

4.3.6 Historical Trend. Here, we provide an additional layer of analysis by exploring the temporal trends of these aspects from 1981 to 2024. This temporal analysis aims to reveal the progression of research priorities and highlight pivotal moments of technological advancement.

As shown in Figure 8, navigation has historically dominated the research landscape, but other application areas have emerged

recently. This perhaps reflects the trends in accessibility and HCI broadly - the research community has begun to understand the challenges in the lives of BLV users more holistically. For example, recent publications have explored challenges for BLV users in daily tasks such as doing make-up [76], cooking [75], or playing video games [45], and have started to design and develop assistive technologies for these problems.

4.4 Embodiments

This section is devoted to the discussion of common physical attributes of these robots. Specifically, we categorized the robots into Robotic Canes, Quadruped & Guide Dog Robots, Drones, and Desktop Robots.

4.4.1 Robotic Canes. BLV people have used white canes as mobility tools for centuries, primarily to detect obstacles and changes to the terrain while walking. The development of robotic canes, therefore, has naturally attracted attention from researchers. Among the papers we examined, 10 (13%) papers are related to the design and development of robotic canes. Robotic canes are typically equipped with a comprehensive set of sensors, such as Lidar, cameras, and Inertial Measurement Units (IMUs, usually include accelerometers to sense motion and gyroscopes to sense orientation), to perceive the environment better.

One example of recent endeavors is Augmented Cane [122], which maintains the form factor of a traditional cane but includes intelligent functionalities like obstacle avoidance and wayfinding. Based on the user study in both indoor and outdoor scenarios, the Augmented Cane can significantly increase walking speed for the users. However, some drawbacks that are reported constantly include weight, durability, and price [94, 122].

4.4.2 Quadruped & Guide Dog Robots. Guide dogs are important for BLV users, both as companions and mobility aids; as mobility aids, they provide assistance in both micro-navigation and macro-navigation (to familiar places). In this vein, we also found 10 (13%) papers on guide dog robots. One of the earliest guide dog robots being built is MELDOG. The authors identified obedience in navigating or guiding a BLV user, as well as intelligent disobedience in detecting and avoiding obstacles, as the key requirements [127, 128]. Due to the hardware and software limitations of the time, however, these prototypes were bulkier and less intelligent than real guide dogs.

In recent years, quadruped robots (which often resemble guide dogs for their four legs), have been explored extensively as robotic guide dogs [57]. Nowadays, it is often a long and intense process for BLV people to obtain a guide dog, due to the scarcity of guide dogs worldwide [55]. Hence, guide dog robots can potentially transform the lives of BLV users. However, there are significant challenges that need to be overcome. For example, surprisingly, a recent study with BLV users revealed that commercial quadruped robots exposed significant disadvantages in usability and trust compared with wheeled robots [135]. They commented that the empathetic effect of the robot’s bionic form was less important compared to its functionality.

4.4.3 Drones. Unmanned Aerial Vehicles (widely known as drones) have become increasingly common in civilian use cases [12, 52]. In



Figure 9: Diverse embodiments of assistive robots for BLV users. A) Robotic Cane, B) Quadruped Robot, C) Drone, D) Suitcase-shaped robot, E) Desktop robot, F) Mechanical bot. Courtesy to the authors.

this review, we identified 8 (10%) papers on drone-based assistive devices for BLV users. Drones offer distinct advantages due to their flying skills and bird-eye view (when equipped with a camera). One notable example is DroneNavigator, where the study explored using both leashed and unleashed drones to lead BLV users in unfamiliar indoor environments [7]. Relatedly, others have explored using drones to assist BLV users in other contexts, such as doing sports [78] or locating objects [56].

A previous survey showed that BLV users envision drones as versatile assistants (e.g., exploring new urban and natural places). They also proposed different modes of piloting that make the flying experience more accessible [44]. However, using drones in civilian spaces raises significant societal and ethical limitations, particularly in privacy and safety [136]. To date, some jurisdictions restrict drones from entering public spaces [34].

4.4.4 Wheeled Grounded Robots. We identified a huge body of literature on developing wheeled grounded robotic systems, with 27 (36%) papers in total¹¹. This includes walker & rollator robots, suitcase robots, shopping cart robots, and other grounded robots (e.g., Pioneer 2DX platform), and they are mainly used for navigation (both micro-navigation and macro-navigation). Most of them leveraged familiar mobile platforms as undercarriages. For example, Cabot is a suitcase-shaped robot with a set of sensors that can help BLV users in navigation by providing vibro-tactile (via the suitcase handle) and audio feedback [49]. The suitcase shape was chosen since the authors believed it could help the BLV users blend with the surroundings without attracting too much attention. A significant limitation of this form factor, as noted in the literature, is its inability to navigate diverse terrains (e.g., curbs, stairs) effectively. [26, 64].

4.4.5 Desktop Robots. We found 15 (20%) papers that investigated desktop robots. These studies usually used commercially available models such as Dash¹², Cellulo¹³, and Ozobots¹⁴. These robots

¹¹Note that these are also referred to as mobile robots in the literature.

¹²<https://www.makewonder.com/en/dash/>

¹³<https://www.epfl.ch/labs/chili/index-html/research/cellulo/>

¹⁴<https://ozobot.com/>

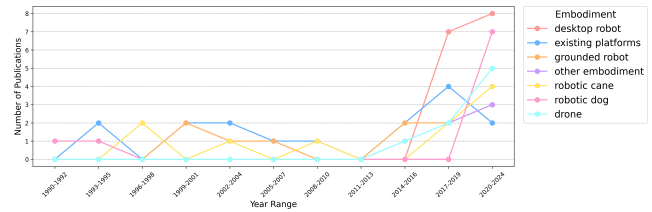


Figure 10: Historical Trend on Embodiments

are affordable and interactive, making them well-suited for classroom settings. Researchers have expanded the application of these robots to support children with visual impairments, facilitating inclusive and playful activities [113], group discussions [99], and games storytelling [6].

4.4.6 Historical Trend. As shown in Figure 10, we observe that some embodiments have gained popularity in recent years, especially robotic guide dogs (quadruped robots), desktop robots, and drones. The latter two embodiments only appeared around five years ago, but they have quickly become an integral part of the research agenda. Traditionally, all other embodiments were built upon or inspired by existing assistive technologies (e.g., white canes, guide dogs), and were fundamentally terrestrial (i.e., operate solely on the ground). This trend indicates how such assistive robots have expanded into new spaces in human society (e.g., in the air). We believe this marks a moment for HCI researchers and practitioners to both celebrate and reflect. On the one hand, assistive robots have become more versatile and powerful; on the other hand, it is important to design and develop these robots carefully so that they can integrate seamlessly into our society, considering their societal and ethical impacts on not only BLV users but also the broader public.

4.5 Input Mechanisms

Throughout this section, we use the term “*explicit*” to refer to the overt input and via command-based or graphical user interface-based interfaces; in contrast, we use the term “*implicit*” to refer to the activity that occurs without the explicit behest or awareness of the user [59].

4.5.1 Explicit Input. We identified 58 (76%) papers that incorporated explicit input. Three predominant types of explicit input mechanisms emerged: haptic input through direct interaction (e.g., pressing buttons and steering handles of the robots), haptic input via external interfaces (e.g., smartphone), and speech input. Direct interaction with robots via built-in interfaces is the most common input method, with 33 (43%) papers falling into this category. These interfaces typically involve touched-based actions such as pressing buttons and keypads [68], as well as steering handles or harnesses [31, 70]. For BLV users, these options allow them to control different aspects of robots’ behavior, such as speed and direction, accurately. Alternatively, external interfaces that are not attached to the robotic systems, such as smartphones for in-situ interactions [64, 77], and tangible blocks [114] in educational settings, are also proposed. We found that 16 (23%) papers have utilized these interfaces. These solutions offer greater flexibility, unrestricted by the physical proximity of the robot.

Finally, speech (particularly verbal commands), has emerged as an alternative paradigm for interacting with robots. Our review identified 11 (14%) papers that envisioned or implemented speech as an input channel. This approach allows users to instruct robots intuitively and naturally. One recent example is Dragon [80], where the assistive robot is powered by a natural dialogue system. In fact, past research has shown that verbal communication is one of the most preferred input modalities when interacting with assistive robots [13, 44], and verbal communication improves perceptions of the social presence of robots [48].

4.5.2 Implicit Input. Conversely, our review identified 7 papers (9%) incorporating implicit inputs, which include user behaviors such as walking speed and gait patterns. The resulting assistive robot can automatically adjust its speed [104] or the tension of the connected elastic rope [31] to match the user’s pace.

Traditionally, HCI researchers have focused on explicit interaction (e.g., graphical user interface-based paradigm in computers). As interactions increasingly occur in everyday life beyond traditional computing environments, where users may be physically and cognitively occupied, implicit interaction has shown promising results in making technology “smarter” [59]. In the realm of robotics, there are some pioneering works on implicit interfaces (e.g., affective cues) to create more seamless and efficient human-robot communication [86, 97]. This shift is particularly significant for BLV users, as listening to ambient sounds or using a white cane are both cognitively and physically demanding tasks.

4.6 Output Mechanisms

4.6.1 Audio. Hearing is undoubtedly an important sensory channel for BLV users, making audio a natural choice for interaction modality. We identified 42 (59%) papers that incorporated audio feedback. Among these papers, 14 (18%) papers incorporated non-verbal cues (e.g., beep sound), and 35 (46%) papers incorporated verbal feedback (e.g., synthesized speech). Most verbal outputs are used as accurate and intuitive feedforward mechanisms to communicate the robots’ intentions or other information to the users, whereas non-verbal cues are used as an additional layer of output, usually for alerting. For example, Kayukawa et al. have proposed a robotic system to help BLV users walk more seamlessly with nearby pedestrians in public spaces by using both beep sounds and synthesized speech to alert them [63].

4.6.2 Haptic. Our analysis revealed that there is a huge body of work using haptic feedback, with 54 (71%) papers incorporating haptic feedback of various kinds. Inspired by previous work [73, 118], we further categorized haptic perception into *cutaneous* and *kinesthetic* feedback. Cutaneous feedback uses sensory information derived from the receptors embedded beneath the skin (e.g., pressure), whereas kinesthetic feedback uses sensory information from mechanoreceptors in muscles, tendons, and joints, thus usually related to the movement and position of the body and its parts.

We found that 32 (42%) papers incorporated kinesthetic feedback and 28 (37%) papers incorporated cutaneous feedback. We encountered a variety of feedback types. For example, cutaneous feedback includes vibrations [134], tactile (e.g., Braille display [69], shapes of the robots [91]), and electrocutaneous stimulation (which refers

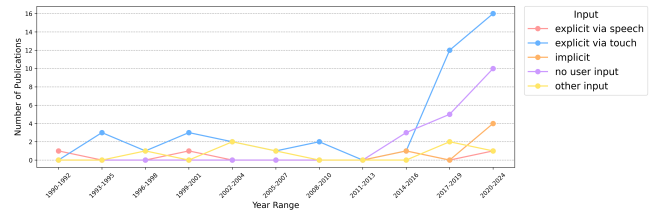


Figure 11: Historical Trend on Input Methods

to the use of electrical currents to stimulate the skin) [128]. Haptic feedback can be exerted on the hand through the harness, leash, or handle of the robotic system; and it can also be applied to other body parts, such as the wrist [104], waist [46], and leg [79].

4.6.3 Others. Apart from the two primary interaction modalities mentioned earlier, we also encountered some less frequently used modalities. One example is movement/proxemics, which have been mentioned in 15 (20%) papers. In this context, movement refers to the ability of robots to change their position within a space to interact with users. This modality is often employed in desktop robots. For instance, these small robots can be used to facilitate group discussions among children with mixed visual abilities by moving closer to the least participative children - to encourage them to speak up [99].

Another example is visual feedback (e.g., LED lights and colorful decorations), which was found in 6 (8%) instances of the reviewed papers. While visual feedback may seem counterintuitive for BLV users, it can still be helpful, particularly for those who have residual vision or perception of light. Lastly, we found 3 (4%) instances where user interaction with the robot was partially or fully mediated through smartphones.

4.6.4 Historical Trend. As shown in Figure 11, we observe that most assistive robots relied primarily on touch-based interfaces or provided no input channels for BLV users. While many BLV participants express a preference for speech-based input in various studies [9, 44], the development of speech-based interfaces remains scarce. This gap likely stems from technical challenges, as speech input systems are susceptible to errors and ambiguity [120]. The data also shows an emerging trend in the exploration of implicit input methods in recent years, which can potentially open up more discussions on “smarter” and more user-centric assistive robots for BLV users.

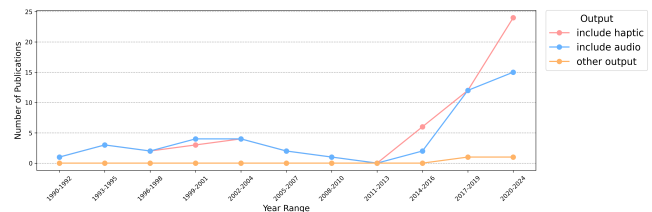


Figure 12: Historical Trend on Output Methods

Remarkably, we observe that haptic and audio feedback are the most common output modalities, with haptic feedback showing

a slight increase in popularity in recent years. This trend may be attributed to the inherent physical contact between robots and users. However, both modalities have limitations. Audio feedback, whether verbal or non-verbal, can be overwhelming and even dangerous for BLV users because they depend on their hearing for ambient sounds. While haptic feedback offers an alternative, it can only encode a limited amount of spatial and temporal information compared to auditory or visual feedback [73]. Further, BLV users often need to exert cognitive effort to interpret haptic information (refer to Section 4.8 for more feedback and insights from BLV users).

4.7 Design & Evaluation Methods

Stepping away from the robotic systems, this section is devoted to research methods. In particular, we pay attention to the commonly used practices, as well as methodological challenges associated with design, evaluation, and participant interactions.

4.7.1 Design. Participatory design and co-design are both well-established methods with a rich history in HCI [66, 123]. In the context of accessibility, involving people with disabilities in the design and development process can help researchers integrate their domain knowledge with the *tacit* knowledge from target users. In contrast with domain knowledge (e.g., prototyping skills), tacit knowledge is often difficult to formalize or describe otherwise (e.g., the living experience of BLV people). Therefore, participatory design is consistently recommended to create assistive technologies that are truly meaningful for the target users [25, 85].

In our analysis, we identified only a small proportion of papers that have involved BLV users in the design process. Specifically, 17 (22%) papers explicitly mentioned the use of design methods such as co-design or participatory design (e.g., focus group discussions), while an additional 8 (11%) papers mentioned preliminary discussions, consultations, or pilot studies involving BLV users or other stakeholders (e.g., Orientation & Mobility instructors).

4.7.2 Evaluation. We identified 49 (64%) papers that conducted user studies, while 7 (9%) papers evaluated their systems using alternative methods such as online surveys or evaluation workshops. The remaining papers provided minimal or no details about their user evaluation process. Among the papers that included user studies, 32 (42%) incorporated experimental studies comparing different conditions or baselines (e.g., compare the drone-based system with an audio-based system [56], or compare the suitcase robot with a mobile app [70]).

Our analysis also revealed some common evaluation measures. We found that task performance metrics (e.g., task completion time and number of collisions) are frequently used as *objective* measures. Conversely, researchers often use the Likert scale or open-ended questions to gather *subjective* insights from users about their experiences with the robotic systems. In Section 4.8, we will delve deeper into the insights and feedback provided by the BLV users.

However, limitations exist across the studies. First, 47 (62%) studies were conducted in research labs that differ from real-world contexts in which the assistive robots were intended to be used. Although lab studies offer greater control, real-world environments offer greater ecological validity. Furthermore, longitudinal studies

are rare. Only 3 (4%) papers have stated that the same group of participants have interacted with the robots for at least two sessions. These findings amplify concerns from previous work [85]. Lastly, while objective measures are important to accurately quantify the performance of assistive robots, complementing them with subjective feedback from BLV users can provide a more holistic understanding, especially *qualitative* feedback. We found that the qualitative component, however, is usually less rigorous. In total, 44 (58%) papers have provided qualitative feedback from the participants (e.g., verbatims and quotes) but only 18 (24%) of the papers amongst them have reported using a systematic method (e.g., thematic analysis [20] and grounded theory [131]) to analyze qualitative data.

4.7.3 Participants. The demographic and number of participants are also key components in the evaluation. The majority of the papers presented at least one user study or pilot study, with only 7 (9%) did not involve any users. We examined the profiles of the participants in all the remaining studies, focusing on their visual abilities. We found 47 (62%) papers included BLV participants in some stages of the evaluation process, and 2 (3%) included only appropriate proxy users (e.g., educators of the BLV students).

Surprisingly, 20 (26%) papers either included only blindfolded sighted participants or did not report on the visual conditions of the participants. While these papers described the design and development of assistive robots for BLV users, it is concerning given the lack of insights and feedback from the BLV users. While we understand it is challenging to recruit representative participants (in our case, BLV users) in research settings, sighted participants lack the same experiences and knowledge as BLV users [115]. Therefore, we argue including only sighted users can lead to inaccurate conclusions or missed insights, which echoes the findings from previous literature [25, 115].

Turning to the numbers, the average number of BLV participants included in the final evaluation is 10.5 (median=8, standard deviation=11).¹⁵ The high variance suggests that there is a considerable variation in the sample sizes across different studies. For example, the usability and performance of the NavCane were tested with 80 BLV participants [90], making it one of the studies with the most users among the papers we reviewed. We will discuss more on the best practices for working with BLV users in Section 5.

4.8 Participants' Insights and Feedback

This section presents feedback and insights from BLV participants. We begin with quantitative feedback, followed by a discussion of recurring themes in the qualitative feedback on assistive robots.

4.8.1 Quantitative Feedback. As we mentioned earlier, 31 (41%) papers used Likert scale questions or other questionnaires (e.g., SUS [24], QUEST [37]) to gather subjective feedback on usability. One common questionnaire is the System Usability Scale (SUS), which is a popular measurement of perceived usability [24]. We found 9 (12%) papers using this scale, and the average SUS score across the

¹⁵Note that this calculation only considers the number of participants in the final evaluation and does not account for any participants involved in earlier stages of the design process or pilot studies.

studies with available SUS score is 75 (standard deviation = 12.8)¹⁶. While we should not make absolute judgments on the “goodness” of the proposed systems based on SUS scores, this constitutes an acceptable SUS score [10].

Interestingly, several studies reported cases where the proposed systems, which covered various robot embodiments, failed to outperform baseline conditions based on feedback from the BLV participants. For example, BLV participants rated PathFinder (a suitcase-shaped robot) higher than their regular aids, but lower than mobile phones in terms of confidence and cognitive load based on the Likert scale questions [70]. Similarly, AugmentedCane (a robotic cane) and Rdog (a robotic dog) were rated slightly lower than traditional aids based on SUS score [26, 122]. While we understand that the small sample sizes do not allow for traditional empirical tests, we believe these findings provide valuable indicators that deserve careful consideration by the HCI community.

4.8.2 Qualitative Feedback. Although 44 (58%) included qualitative feedback from BLV participants, only 18 (24%) mentioned systematic analysis methods when reporting them. While qualitative data is limited, several interconnected themes arose by focusing on the critical comments from the BLV participants.

In general, the proposed robots received positive and encouraging feedback from the BLV users. One common issue constantly mentioned by BLV participants is the *weight* and *size* of the assistive robots. For example, nearly all the participants agreed that the AugmentedCane was too heavy [122], making weight one of the system’s worst attributes. Another frequently discussed area for improvement is the interface design. For example, some BLV participants felt the audio interface could interfere with the ambient sounds (which is important for them), and could not work well in noisy environments [63]. In addition, they were also concerned about the *noise* from the robot themselves [56, 135]. Turning to the haptic interface, a common issue reported being the *learning curve* to understand the haptic patterns [103, 122].

Noticeably, we found that BLV participants often hold mixed views on these topics, depending on their vision abilities and the specific tasks (e.g., unfamiliar or familiar, indoor or outdoor). For instance, two studies concluded that BLV participants exhibited different preferences on *autonomy* [109, 141] when choosing between full and partial control. Perspectives on design priorities also varied significantly: while some emphasized functional capabilities over aesthetic consideration [135], others expressed strong concerns about social acceptance and appearance [49, 61]. Some BLV users concluded that assistive robots should offer customization options to accommodate individual needs and preferences [9]. Further studies are also needed to explore these diverse preferences and understand how they might influence the adoption and long-term use of assistive robots.

¹⁶In studies where multiple modes were identified, the interpretation depends on the research objective: when selecting the optimal mode, the highest value is considered; when examining overall usability, all possible modes are taken into account.

5 DISCUSSION

5.1 RQ1: What are the research opportunities in Human-Robot Interaction for BLV users?

This section is devoted to potential avenues for future research. These directions are informed by the contradictions, gaps, and challenges we identified in our surveyed body of work, especially when positioning them within the greater discourse of accessibility and assistive technology. For each theme, we first describe the current state and research gap. Then, we provide potential directions and their relevant works, which we hope to serve as sources of inspiration for future HCI researchers and practitioners.

5.1.1 Theme 1: Customization and Adaptability.

Status-quo and Gaps: We found that most assistive robots are designed for “general” blind users, when there is no such thing as a general blind user because everyone BLV user is unique (refer to Figure 7). Many BLV participants have mentioned the benefits of customization (refer to 4.8), highlighting that a one-size-fits-all approach usually falls short of adequately addressing their wide spectrum of needs shaped by various factors such as vision abilities, age of onset, and other considerations (e.g., culture) [125]. HCI researchers also advocate for customization for assistive technologies [53, 139]. Note that this concept is sometimes referred to as *adaptability* in the literature, by which it means the degree to which software can be customized for a user [139].

Research Directions and Relevant Literature: First, more research is needed to identify useful and relevant customization options. In the broader discussion of assistive robots and assistive technologies, previous studies have attempted to personalize the autonomy level or the embedded algorithms [96, 141]. We believe exploring these directions will increase the impact of assistive robots for BLV users.

Second, it is important to provide accessible and user-friendly interfaces for BLV users themselves to customize assistive robots (similar to how one might need to configure their smartphones). We would like to highlight the potential of end-user programming paradigms to transform how BLV users might customize and use assistive robots. End-user programming aims to empower non-expert users to instruct and program robots for personal use. While traditional methods often utilize visual and tangible elements (e.g., blocks and toggles) [106], it is exciting to explore natural language as a more intuitive, accessible, and expressive interface for BLV users [60, 80].

5.1.2 Theme 2: Social and Emotional Wellbeing.

Status-quo and Gaps: Assistive robots have demonstrated great potential in offering emotional support (refer to Section 2.1), this is mostly for the elderly who are more likely to face isolation and stigma from their conditions (e.g., dementia). However, when it comes to BLV users, it has yet to be explored. Actually, more than half of the BLV population are above 60 since age-related issues (e.g., macular degeneration, diabetes) are the most common causes of visual impairment [23, 130]. Therefore, we propose expanding the role of assistive robots beyond physical assistance to provide meaningful emotional support for BLV users, especially elderly BLV.

Research Directions and Relevant Literature: First, we believe it is exciting to explore robot companions for BLV users. Robots have been explored as companions for humans in various activities, including eating [102], shopping [14], and reading [93]. In the context of elderly care, robots have also been successfully employed as companions in their daily lives to support social-emotional well-being.

Researchers could also leverage the metaphor of “*pet*”, as we have seen in previous assistive robots such as AIBO and PARO (refer to Figure 2). Similarly, the design of robotic guide dogs for BLV users has been inspired by their biological counterparts. Notably, guide dogs provide not only assistance but also social and psychological benefits, which directly and indirectly improve their owners’ quality of life [138]. It has been well-evidenced that adding *anthropomorphic* (i.e., human-like) or *zoomorphic* (i.e., animal-like) features and behaviors can foster meaningful social connections between users and robots [43]; most studies in the past used visual features (e.g., face, eye) to elicit familiarity and likeability, but we believe additional modalities such as affective touch (tactile feedback with a hedonic or emotional component) [108] or voice interfaces [28] are more suitable for BLV users.

5.2 RQ2: What are the key considerations in designing assistive robots for BLV users?

In this section, we underpin key considerations for designing assistive robots that support BLV users. We discuss these considerations around three axes: users, environments, and other stakeholders.

5.2.1 Interacting with Users: Balancing Robot Autonomy and User Agency. We believe that Human-Centered AI principles, such as those proposed by Schneidermann et al. [120] and Microsoft [5], are essential for designing the functionalities and interfaces of assistive robots. As assistive robots become more intelligent, developing trustworthy and meaningful assistive robots requires a delicate balance between robot intelligence and user agency. On the one hand, this balance can increase user trust and robotic system usability over time; on the other hand, it is crucial to preserve human control during anomalous situations [120].

In the context of supporting BLV users, it is important to understand the needs and capabilities of the users first. For instance, who should make the decision and what information is needed for them to make the most informed decision. Second, it is also important to provide accessible interfaces for BLV users to modify and override robot decisions, so that they are *in control of* rather than *controlled* by the robotic device. This agrees with previous studies that examined BLV users’ preference for autonomy and control when interacting with assistive robots [3, 109, 141].

5.2.2 Interacting with Environments: Long-term Deployment in Real-world Environments. Assistive robots are difficult to develop and distribute, they are also expensive to maintain and support. Throughout the analysis, we discovered several technical limitations of current prototypes, which are typically acknowledged by the authors themselves. In terms of hardware, commonly reported issues include battery life, camera functionality, and overall weight. For example, Hwang et al. suggested that the batteries

of guide dog robots should last at least two hours [57]. Software-related issues are equally critical in real-world applications of assistive robots for BLV users. Currently, one significant challenge in software development for these robots is the *generalizability* and *robustness* of algorithms in dynamic environments. For instance, intersection detection algorithms must recognize diverse intersection types across various contexts [70], and navigation systems need to function effectively in crowded spaces [64].

These limitations hinder the deployment of robots in real-world settings (refer to Section 4.7.2). While lab-based only evaluations draw limited insights from BLV participants, that may lack depth and authenticity, given that lab-based simulations cannot fabricate real-world situations. Thus, we believe future work must consider these challenges when designing and developing assistive robots for BLV users.

5.2.3 Interacting with Other Stakeholders: Social Acceptance. When assistive robots enter real-world environments from laboratories, they will transition from a closed system to an *open system* with numerous events and people, interacting with people outside BLV individuals in real-world [120]. Therefore, to seamlessly integrate assistive robots into society, it is critical to consider all stakeholders in the ecosystem, including bystanders or caretakers. For example, as robots move closer to people, ethical and safety considerations become more pressing [8]. Furthermore, *aesthetics* and *social acceptance* of assistive technologies hold importance for BLV users, as many of them view these technologies as part of their identities [74, 119].

In this vein, some studies explored suitcase-shaped robots, because the authors believe this embodiment can help BLV users blend into their surroundings without drawing too much attention [49]. Relatedly, preliminary results have also suggested that most of the public has a positive attitude towards visibly identified assistive robots [8]. However, a rich design space for assistive robot interactions with the public still needs to be explored further.

5.3 RQ3: What are the challenges in conducting research in this field, and how should we approach them?

We observed various practices and methods employed, which mirrors the findings from previous literature [25]. While we believe there is no single valid approach to conducting research in this field given the interdisciplinary nature of HCI research, we offer some actionable and concrete recommendations for future research.

5.3.1 Design “with”, Not Design “for” BLV Users. While creating novel assistive technologies is important, we contend the usefulness and meaningfulness of the proposed solution are equally important. One challenge is that designers and developers often struggle to fully understand the perspectives of BLV users. Recently, the HCI community called for a shift from designing “*for*” users with disabilities to designing “*with*” them, this way, designers and developers empower BLV users to impart their needs and shape their technological future [25, 85]. Their collaboration is witnessed in various stages of participatory design and co-design methods. For instance, interviews, focus group discussions, and pilot studies [123].

Notably, there are some emerging concerns to how these methods are performed and communicated [11, 132]. When it comes to user studies, for example, tech-savvy BLV users who are also fluent communicators are more likely to be selected; but this group might not be representative participants for more marginalized groups (e.g., BLV users with multiple intersecting conditions). In light of this, other variations such as situated participatory design have been proposed, which situates the activity in a genuine and realistic environment [124].

5.3.2 Gather Feedback from BLV Social Network. Previous surveys have highlighted that recruiting users with disabilities in research settings is challenging [115]. According to Lazar et al., it is difficult to obtain beyond 5-10 users per study [72]. We found some studies that also chose to engage with “proxy users”, such as educators, family, and caregivers of BLV users [91, 114]. We believe it is appropriate, as they often have a considerable amount of understanding of the experiences of BLV users, and may also interact with the robots themselves if these assistive robots are made available, which resonated in previous literature [25, 72]. Nevertheless, we would like to emphasize that BLV users perceive and interact with robots in ways that are fundamentally different from sighted users due to their unique needs and preferences.

5.3.3 Use Wizard-of-Oz Prototyping to Elicit Early Feedback. The Wizard-of-Oz (WoZ) has a long tradition within the field of Human-Robot Interaction. It refers to the experimenter remotely controlling any number of aspects of the robot [65]. Of our surveyed body of work, WoZ has been used as well [9, 105], though relatively limited. We believe, however, that it is particularly useful when the robots are not currently advanced enough to interact autonomously with BLV participants in socially appropriate or safe ways, and the WoZ method allows BLV participants to envision what future interaction could be like.

However, it is important to employ the WoZ method in a rigorous and repeatable manner, thus allowing for a smoother transition to a more autonomous and capable system in the future - it is not intended to be the end point of development, but rather a starting point of an iterative human-centered process. We also recommend HCI readers turn for guidance and inspiration from relevant literature in the broader discussion in HRI [54, 112].

5.3.4 Collect Rich and Specific Qualitative Feedback. Lastly, rich and specific qualitative feedback can help researchers to better understand BLV users’ perspectives, for example, explanations of why a system works or fails. It is valuable especially when working with a niche population such as BLV users, where small sample sizes do not allow for traditional statistical tests (refer to Section 4.8). However, qualitative analysis needs to be *reliable* and *valid*.

We observe that the analysis of qualitative data often lacks the same level of rigor as that applied to quantitative data, as only a small portion of the papers with qualitative data have reported using a systematic method of analysis (refer to Section 4.7.2). Due to the unstructured nature of qualitative data, it is challenging to analyze it systematically. In light of this, we recommend that researchers employ established qualitative analysis methods, such as thematic analysis [20], reflexive thematic analysis [22], and grounded theory [131] to interpret qualitative data. We would also like to point

HCI researchers to some relevant literature [137] to better understand the different worldviews that inform both quantitative and qualitative research (e.g., Positivist, Interpretive).

6 LIMITATIONS

This review has a couple of limitations. First, we decided to analyze and synthesize the papers through an HCI lens and provide a holistic understanding of the current landscape. Nevertheless, we acknowledge that other researchers could approach this problem differently. As such, we would like to point out that other dimensions of data could be coded. For instance, future researchers could investigate the types of commonly used technologies of these robotic systems, which could enrich the understanding of the field beyond the scope of the current study.

In addition, curating a literature survey dataset is challenging. This is largely because the selection of the papers involves a substantial manual effort to concretize the barriers, which are prone to errors and bias. Lastly, we have only focused on the Scopus database and certain venues in this review. Admittedly, even though we have applied various strategies to ensure the reliability and rigor of our study (e.g., calculating IRR, and carefully defining inclusion/exclusion criteria), we may have missed some relevant papers.

7 CONCLUSION

In this paper, we conducted a systematic literature review to investigate how robots can be used to support BLV users in different contexts and tasks. We identified a final corpus of 76 papers from premier HCI venues on this topic. Through our structured analysis, we examined the key elements of these robotic systems, including their application areas, embodiments, and interaction techniques. We also reported on the study methods and practices that researchers have chosen to design and develop these robots.

More importantly, our findings shed light on gaps and challenges in this domain. For example, while there has been a growing interest in this area in recent years, we observed that most studies predominantly focused on navigation-related issues. Furthermore, our analysis revealed a critical need for greater engagement with BLV users throughout the design process to better understand and incorporate their needs and preferences. We hope future work can use our data and ideas to develop a broader range of robotic systems that can genuinely enhance the quality of life for BLV users.

8 ACKNOWLEDGEMENT

This work was supported by the Ministry of Education AcRF Tier 1 Grant 251RES2208. We would like to thank Augmented Human Lab (<https://ahlab.org/>) as well as Cherchons pour Voir Lab (<https://cherchonspourvoir.org/>) for the support. Specifically, we would like to thank Yadeesha Weerasinghe for helping with the project website and Hanjun Michelle Wu for proofreading. Lastly, we would like to thank the reviewers for their invaluable feedback.

REFERENCES

- [1] Jordan Abdi, Ahmed Al-Hindawi, Tiffany Ng, and Marcela P Vizcaychipi. 2018. Scoping review on the use of socially assistive robot technology in elderly care. *BMJ Open* 8, 2 (2018), e018815.
- [2] Shivendra Agrawal, Mary Etta West, and Bradley Hayes. 2022. A novel perceptive robotic cane with haptic navigation for enabling vision-independent participation in the social dynamics of seat choice. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (Kyoto, Japan). IEEE, IEEE, 9156–9163.
- [3] Peter Aigner and Brenan McCarragher. 1999. Shared control framework applied to a robotic aid for the blind. *IEEE Control Systems Magazine* 19, 2 (1999), 40–46.
- [4] Morana Alač. 2016. Social robots: Things or agents? *AI & Society* 31 (2016), 519–535.
- [5] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland, UK). ACM, Article 3, 13 pages.
- [6] Cristiana Antunes, Isabel Neto, Filipa Correia, Ana Paiva, and Hugo Nicolau. 2022. Inclusive'R'Stories: An Inclusive Storytelling Activity with an Emotional Robot. In *2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (Sapporo, Hokkaido, Japan). IEEE, 90–100.
- [7] Mauro Avila, Markus Funk, and Niels Henze. 2015. Dronenavigator: Using drones for navigating visually impaired persons. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers and Accessibility* (Lisbon, Portugal). ACM, 327–328.
- [8] Mauro Avila Soto and Markus Funk. 2018. Look, a guidance drone! assessing the social acceptability of companion drones for blind travelers in public spaces. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility* (Galway, Ireland). ACM, 417–419.
- [9] Shiri Azenkot, Catherine Feng, and Maya Cakmak. 2016. Enabling building service robots to guide blind people a participatory design approach. In *2016 11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 3–10.
- [10] Aaron Bangor, Philip T Kortum, and James T Miller. 2008. An empirical evaluation of the system usability scale. *International Journal of Human-Computer Interaction* 24, 6 (2008), 574–594.
- [11] Liam Bannon, Jeffrey Bardzell, and Susanne Bødker. 2018. Introduction: Reimagining participatory design—Emerging voices. , 8 pages.
- [12] Mehmet Aydin Baytas, Damla Çay, Yuchong Zhang, Mohammad Obaid, Asim Evren Yantaç, and Morten Fjeld. 2019. The design of social drones: A review of studies on autonomous flyers in inhabited environments. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Article 250, 13 pages.
- [13] Jenay M Beer, Akanksha Prakash, Cory-Ann Smarr, Tracy L Mitzner, Charles C Kemp, and Wendy A Rogers. 2012. “Commanding your robot” older adults’ preferences for methods of robot control. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, Vol. 56. SAGE Publications Sage CA: Los Angeles, CA, 1263–1267.
- [14] Francesca Bertacchini, Eleonora Bilotta, and Pietro Pantano. 2017. Shopping with a robotic companion. *Computers in Human Behavior* 77 (2017), 382–395.
- [15] Tapomayukh Bhattacharjee, Gilwoo Lee, Hanjun Song, and Siddhartha S Srinivasa. 2019. Towards robotic feeding: Role of haptics in fork-based food manipulation. *IEEE Robotics and Automation Letters* 4, 2 (2019), 1485–1492.
- [16] Júlia Pareto Boada, Begoña Román Maestre, and Carme Torras Genis. 2021. The ethical issues of social assistive robotics: A critical literature review. *Technology in Society* 67 (2021), 101726.
- [17] Roger Boldu, Alexandru Dancu, Denys JC Matthies, Thisum Buddhika, Shamane Siriwardhana, and Suranga Nanayakkara. 2018. Fingerreader2.0: Designing and evaluating a wearable finger-worn camera to assist people with visual impairments while shopping. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 3 (2018), 1–19.
- [18] Roger Boldu, Denys JC Matthies, Haimo Zhang, and Suranga Nanayakkara. 2020. AiSee: an assistive wearable device to support visually impaired grocery shoppers. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–25.
- [19] Johann Borenstein and Iwan Ulrich. 1997. The guidecane—a computerized travel aid for the active guidance of blind pedestrians. In *Proceedings of International Conference on Robotics and Automation*, Vol. 2. IEEE, 1283–1288.
- [20] Robert Bowman, Camille Nadal, Kellie Morrissey, Anja Thieme, and Gavin Doherty. 2023. Using thematic analysis in healthcare HCI at CHI: A scoping review. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 491, 18 pages.
- [21] Erin Brady, Meredith Ringel Morris, Yu Zhong, Samuel White, and Jeffrey P Bigham. 2013. Visual challenges in the everyday lives of blind people. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2117–2126.
- [22] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health* 11, 4 (2019), 589–597.
- [23] Neil M Bressler. 2004. Age-related Macular Degeneration is the Leading Cause of Blindness. *JAMA* 291, 15 (2004), 1900–1901.
- [24] J Brooke. 1996. SUS: A quick and dirty usability scale. *Usability Evaluation in Industry* (1996).
- [25] Emeline Brulé, Brianna J Tomlinson, Oussama Metatla, Christophe Jouffrais, and Marcos Serrano. 2020. Review of quantitative empirical evaluations of technology for people with visual impairments. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). ACM, 1–14.
- [26] Shaojun Cai, Ashwin Ram, Zhengtai Gou, Mohd Alqama Wasim Shaikh, Yu-An Chen, Yingjia Wan, Kotaro Hara, Shengdong Zhao, and David Hsu. 2024. Navigating Real-World Challenges: A Quadruped Robot Guiding System for Visually Impaired People in Diverse Environments. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). Article 44, 18 pages.
- [27] Sandra Cano, Carina S González, Rosa María Gil-Iranzo, and Sergio Albiol-Pérez. 2021. Affective communication for socially assistive robots (sars) for children with autism spectrum disorder: A systematic review. *Sensors* 21, 15 (2021), 5166.
- [28] Sam WT Chan, Tamil Selvan Gunasekaran, Yun Suen Pai, Haimo Zhang, and Suranga Nanayakkara. 2021. KinVoices: Using voices of friends and family in voice interfaces. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–25.
- [29] Hsuan-Eng Chen, Yi-Ying Lin, Chien-Hsing Chen, and I-Fang Wang. 2015. BlindNavi: A navigation app for the visually impaired smartphone user. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*. 19–24.
- [30] Tiffany L Chen, Matei Ciocarlie, Steve Cousins, Phillip M Grice, Kelsey Hawkins, Kaijen Hsiao, Charles C Kemp, Chih-Hung King, Daniel A Lazewatsky, Adam E Leeper, et al. 2013. Robots for humanity: using assistive robotics to empower people with disabilities. *IEEE Robotics & Automation Magazine* 20, 1 (2013), 30–39.
- [31] Yanbo Chen, Zhengzhe Xu, Zhuozhu Jian, Gengpan Tang, Liyunong Yang, Anxing Xiao, Xueqian Wang, and Bin Liang. 2023. Quadruped guidance robot for the visually impaired: A comfort-based approach. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 12078–12084.
- [32] Tzu-Kuan Chuang, Ni-Ching Lin, Jih-Shi Chen, Chen-Hao Hung, Yi-Wei Huang, Chunhui Teng, Haikun Huang, Lap-Fai Yu, Laura Giarre, and Hsueh-Cheng Wang. 2018. Deep trail-following robotic guide dog in pedestrian environments for people who are blind and visually impaired-learning from virtual and real worlds. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 5849–5855.
- [33] John A Collins and Bart CJM Fauser. 2005. Balancing the strengths of systematic and narrative reviews. *Human Reproduction Update* 11, 2 (2005), 103–104.
- [34] Arthur P Cracknell. 2017. UAVs: regulations and law enforcement. *International Journal of Remote Sensing* 38, 8–10 (2017), 3054–3067.
- [35] Maitraye Das, Darren Gergle, and Anne Marie Piper. 2023. Symphony: Enhancing Accessible Pattern Design Practices among Blind Weavers. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [36] Kerstin Dautenhahn and Aude Billard. 1999. Bringing up robots or—the psychology of socially intelligent robots: From theory to implementation. In *Proceedings of the Third Annual Conference on Autonomous Agents*. 366–367.
- [37] Louise Demers, Rhoda Weiss-Lambrou, and Bernadette Ska. 1996. Development of the Quebec user evaluation of satisfaction with assistive technology (QUEST). *Assistive Technology* 8, 1 (1996), 3–13.
- [38] Giuseppe Desolda, Lauren S Ferro, Andrea Marrella, Tiziana Catarci, and Maria Francesca Costabile. 2021. Human factors in phishing attacks: a systematic literature review. *ACM Computing Surveys (CSUR)* 54, 8 (2021), 1–35.
- [39] Qicheng Ding, Jiangtao Gong, Jingwei Sun, Penghui Xu, Liuxin Zhang, Qianying Wang, and Yu Zhang. 2019. HeliCoach: A drone-based system supporting orientation and mobility training for the visually impaired. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland, UK). ACM, 1–6.
- [40] Julie Ducasse, Marc Macé, Bernard Oriola, and Christophe Jouffrais. 2018. Botmap: Non-visual panning and zooming with an actuated tabletop tangible interface. *ACM Transactions on Computer-Human Interaction (TOCHI)* 25, 4 (2018), 1–42.
- [41] David Feil-Seifer and Maja J Mataric. 2005. Defining socially assistive robotics. In *9th International Conference on Rehabilitation Robotics, 2005. ICORR 2005*. IEEE, 465–468.
- [42] Alexander Fiannaca, Ilias Apostolopoulos, and Eelke Folmer. 2014. Headlock: a wearable navigation aid that helps blind cane users traverse large open spaces. In *Proceedings of the 16th International ACM SIGACCESS Conference on Computers and Accessibility* (Rochester, NY, USA). 19–26.
- [43] Julia Fink. 2012. Anthropomorphism and human likeness in the design of robots and human-robot interaction. In *The 4th International Conference of Social Robotics* (Chengdu, China). Springer, 199–208.

- [44] Vinitha Gadiraju, Jérémie Garcia, Shaun Kane, and Anke M. Brock. 2021. "It is fascinating to make these beasts fly": Understanding Visually Impaired People's Motivations and Needs for Drone Piloting. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, USA). ACM, 1–12.
- [45] David Gonçalves, Manuel Piçarra, Pedro Pais, João Guerreiro, and André Rodrigues. 2023. "My Zelda Cane": Strategies Used by Blind Players to Play Visual-Centric Digital Games. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [46] Jiangtao Gong, Qicheng Ding, Yu Zhang, Liuxin Zhang, Penghui Xu, Mingming Liu, Yingqing Xu, and Qianying Wang. 2020. HeliCoach: An Adaptive Orientation and Mobility Training System in a Drone-Based 3D Audio Space. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). ACM, 1–8.
- [47] Ricardo E Gonzalez Penuela, Jazmin Collins, Cynthia Bennett, and Shiri Azenkot. 2024. Investigating Use Cases of AI-Powered Scene Description Applications for Blind and Low Vision People. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). Article 901, 21 pages.
- [48] Elena Corina Grigore, Andre Pereira, Ian Zhou, David Wang, and Brian Scasselati. 2016. Talk to me: Verbal communication improves perceptions of friendship and social presence in human-robot interaction. In *Intelligent Virtual Agents: 16th International Conference, IVA 2016, Los Angeles, CA, USA, September 20–23, 2016, Proceedings 16*. Springer, 51–63.
- [49] João Guerreiro, Daisuke Sato, Saki Asakawa, Huixu Dong, Kris M Kitani, and Chieko Asakawa. 2019. Cabot: Designing and evaluating an autonomous navigation robot for blind people. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility* (Pittsburgh, PA, USA). 68–82.
- [50] Vicki L Hanson, Anna Cavender, and Shari Trewin. 2015. Writing about accessibility. *Interactions* 22, 6 (2015), 62–65.
- [51] Andrew F Hayes and Klaus Krippendorff. 2007. Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures* 1, 1 (2007), 77–89.
- [52] Viviane Herdel, Lee J Yamin, and Jessica R Cauchard. 2022. Above and beyond: A scoping review of domains and applications for human-drone interaction. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA). ACM, Article 463, 22 pages.
- [53] Jaylin Herskovitz, Andi Xu, Rahaf Alharbi, and Anhong Guo. 2023. Hacking, switching, combining: understanding and supporting DIY assistive technology design by blind people. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 57, 17 pages.
- [54] Guy Hoffman and Xuan Zhao. 2020. A primer for conducting experiments in human-robot interaction. *ACM Transactions on Human-Robot Interaction (THRI)* 10, 1 (2020), 1–31.
- [55] Bin Hong, Zhangxi Lin, Xin Chen, Jing Hou, Shunya Lv, and Zhendong Gao. 2022. Development and application of key technologies for Guide Dog Robot: A systematic literature review. *Robotics and Autonomous Systems* 154 (2022), 104104.
- [56] Felix Huppert, Gerold Hoelzl, and Matthias Kranz. 2021. GuideCopter-A precise drone-based haptic guidance interface for blind or visually impaired people. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan). ACM, Article 218, 14 pages.
- [57] Hochul Hwang, Hee-Tae Jung, Nicholas A Giudice, Joydeep Biswas, Sunghoon Ivan Lee, and Donghyun Kim. 2024. Towards Robotic Companions: Understanding Handler-Guide Dog Interactions for Informed Guide Dog Robot Design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). ACM, Article 596, 20 pages.
- [58] R Dan Jacobson and Robert M Kitchin. 1997. GIS and people with visual impairments or blindness: Exploring the potential for education, orientation, and navigation. *Transactions in GIS* 2, 4 (1997), 315–332.
- [59] Wendy Ju and Larry Leifer. 2008. The design of implicit interactions: Making interactive systems less obnoxious. *Design Issues* 24, 3 (2008), 72–84.
- [60] Ulas Berk Karli, Joo-Tung Chen, Victor Nikhil Antony, and Chien-Ming Huang. 2024. Alchemist: LLM-Aided End-User Development of Robot Applications. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (Boulder, CO, USA). 361–370.
- [61] Seita Kayukawa, Keita Higuchi, João Guerreiro, Shigeo Morishima, Yoichi Sato, Kris Kitani, and Chieko Asakawa. 2019. BBeep: A sonic collision avoidance system for blind travellers and nearby pedestrians. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland, UK). ACM, Article 52, 12 pages.
- [62] Seita Kayukawa, Tatsuya Ishihara, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2020. BlindPilot: A robotic local navigation system that leads blind people to a landmark object. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). ACM, 1–9.
- [63] Seita Kayukawa, Tatsuya Ishihara, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2020. Guiding blind pedestrians in public spaces by understanding walking behavior of nearby pedestrians. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 3 (2020), 1–22.
- [64] Seita Kayukawa, Daisuke Sato, Masayuki Murata, Tatsuya Ishihara, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2023. Enhancing blind visitor's autonomy in a science museum using an autonomous navigation robot. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 541, 14 pages.
- [65] John F Kelley. 1984. An iterative design methodology for user-friendly natural language office information applications. *ACM Transactions on Information Systems (TOIS)* 2, 1 (1984), 26–41.
- [66] Finn Kensing and Jeanette Blomberg. 1998. Participatory design: Issues and concerns. *Computer supported cooperative work (CSCW)* 7 (1998), 167–185.
- [67] Barbara Kitchenham. 2004. Procedures for performing systematic reviews. *Keele, UK, Keele University* 33, 2004 (2004), 1–26.
- [68] Vladimir Kulyukin, Chaitanya Gharpure, and John Nicholson. 2005. Robocart: Toward robot-assisted navigation of grocery stores by the visually impaired. In *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2845–2850.
- [69] Vladimir A Kulyukin and Chaitanya Gharpure. 2006. Ergonomics-for-one in a robotic shopping cart for the blind. In *Proceedings of the 1st ACM SIGCHI/SIGART Conference on Human-Robot Interaction*. ACM, 142–149.
- [70] Masaki Kuribayashi, Tatsuya Ishihara, Daisuke Sato, Jayakorn Vongkulbhisal, Karnik Ram, Seita Kayukawa, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2023. Pathfinder: Designing a map-less navigation system for blind people in unfamiliar buildings. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 41, 16 pages.
- [71] Maria Kyrarini, Manizheh Zand, and Krisha Kodur. 2023. Assistive Robots for Persons with Visual Impairments: Current Research and Open Challenges. In *Proceedings of the 16th International Conference on Pervasive Technologies Related to Assistive Environments*. 413–416.
- [72] Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser. 2017. *Research methods in human-computer interaction*. Morgan Kaufmann.
- [73] Susan J Lederman and Roberta L Klatzky. 2009. Haptic perception: A tutorial. *Attention, Perception, & Psychophysics* 71, 7 (2009), 1439–1459.
- [74] Franklin Mingzhe Li, Di Laura Chen, Mingming Fan, and Khai N Truong. 2021. "I Choose Assistive Devices That Save My Face" A Study on Perceptions of Accessibility and Assistive Technology Use Conducted in China. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan). ACM, Article 372, 14 pages.
- [75] Franklin Mingzhe Li, Michael Xieyang Liu, Shaun K Kane, and Patrick Carrington. 2024. A Contextual Inquiry of People with Vision Impairments in Cooking. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–14.
- [76] Franklin Mingzhe Li, Francesca Spector, Meng Xia, Mina Huh, Peter Cederberg, Yui Gong, Kristen Shinohara, and Patrick Carrington. 2022. "It Feels Like Taking a Gamble": Exploring Perceptions, Practices, and Challenges of Using Makeup and Cosmetics for People with Visual Impairments. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [77] Jiasheng Li, Zeyu Yan, Arush Shah, Jonathan Lazar, and Huaishu Peng. 2023. Toucha11y: Making inaccessible public touchscreens accessible. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 748, 13 pages.
- [78] Zhenyu Liao, Jose V Salazar Lucas, Ankita A Ravankar, and Yasuhisa Hirata. 2023. Running Guidance for Visually Impaired People Using Sensory Augmentation Technology Based Robotic System. *IEEE Robotics and Automation Letters* 8, 9 (Sept. 2023).
- [79] Zhenyu Liao, Jose Salazar, and Yasuhisa Hirata. 2021. Robotic guidance system for visually impaired users running outdoors using haptic feedback. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 8325–8331.
- [80] Shuijing Liu, Aamir Hasan, Kaiwen Hong, Runxuan Wang, Peixin Chang, Zachary Mizrahi, Justin Lin, D Livingston McPherson, Wendy A Rogers, and Katherine Driggs-Campbell. 2024. DRAGON: A dialogue-based robot for assistive navigation with visual language grounding. *IEEE Robotics and Automation Letters* 9, 4 (April 2024), 3712 – 3719.
- [81] Janice Lloyd, Claire Budge, Steve La Grow, and Kevin Stafford. 2016. An investigation of the complexities of successful and unsuccessful guide dog matching and partnerships. *Frontiers in Veterinary Science* 3 (2016), 114.
- [82] Jack M Loomis, Reginald G Golledge, Roberta L Klatzky, and James R Marston. 2020. Assisting wayfinding in visually impaired travelers. In *Applied Spatial Cognition*. Psychology Press, 179–202.
- [83] Ernesto P Lopes, Eliana PL Aude, Julio TC Silveira, and Henrique Serdeira. 2005. Obstacle avoidance strategy based on adaptive potential fields generated by an electronic stick. In *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2626–2631.

- [84] Chen-Lung Lu, Zi-Yan Liu, Jui-Te Huang, Ching-I Huang, Bo-Hui Wang, Yi Chen, Nien-Hsin Wu, Hsueh-Cheng Wang, Laura Giarre, and Pei-Yi Kuo. 2021. Assistive navigation using deep reinforcement learning guiding robot with UWB/voice beacons and semantic feedbacks for blind and visually impaired people. *Frontiers in Robotics and AI* 8 (2021), 654132.
- [85] Kelly Mack, Emma McDonnell, Dhruv Jain, Lucy Lu Wang, Jon E. Froehlich, and Leah Findlater. 2021. What do we mean by “accessibility research”? A literature survey of accessibility papers in CHI and ASSETS from 1994 to 2019. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan). ACM, Article 371, 18 pages.
- [86] Karthik Mahadevan, Mauricio Sousa, Anthony Tang, and Toví Grossman. 2021. “grip-that-there”: An investigation of explicit and implicit task allocation techniques for human-robot collaboration. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan). ACM, Article 215, 14 pages.
- [87] Hamza Mahdi, Sami Alperen Akgun, Shahed Saleh, and Kerstin Dautenhahn. 2022. A survey on the design and evolution of social robots—Past, present and future. *Robotics and Autonomous Systems* (2022), 104193.
- [88] Adriano Mancini, Emanuele Frontoni, and Primo Zingaretti. 2018. Mechatronic system to help visually impaired users during walking and running. *IEEE Transactions on Intelligent Transportation Systems* 19, 2 (2018), 649–660.
- [89] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and inter-rater reliability in qualitative research: Norms and guidelines for CSCW and HCI practice. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–23.
- [90] Vidula V Meshram, Kailas Patil, Vishal A Meshram, and Felix Che Shu. 2019. An astute assistive device for mobility and object recognition for visually impaired people. *IEEE Transactions on Human-Machine Systems* 49, 5 (2019), 449–460.
- [91] Oussama Metatla, Sandra Bardot, Clare Cullen, Marcos Serrano, and Christophe Jouffrais. 2020. Robots for inclusive play: Co-designing an educational game with visually impaired and sighted children. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA). ACM, 1–13.
- [92] Oussama Metatla, Alison Oldfield, Taimur Ahmed, Antonis Vafeas, and Sunny Miglani. 2019. Voice user interfaces in schools: Co-designing for inclusion with visually-impaired and sighted pupils. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland, UK). ACM, Article 378, 15 pages.
- [93] Joseph E Michaelis and Bilge Mutlu. 2017. Someone to read with: Design of and experiences with an in-home learning companion robot for reading. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, 301–312.
- [94] Rezylle Milallos, Vinita Tibdewal, Yiwen Wang, Andre Ogueh Udegbe, and Tae Oh. 2021. “Would the smart cane benefit me?”: Perceptions of the Visually Impaired towards Smart Canes. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, USA). ACM, 1–3.
- [95] Clara Moge, Katherine Wang, and Youngjun Cho. 2022. Shared user interfaces of physiological data: Systematic review of social biofeedback systems and contexts in HCI. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA). ACM, Article 301, 16 pages.
- [96] Cecily Morrison, Martin Grayson, Rita Faia Marques, Daniela Massiceti, Camilla Longden, Linda Wen, and Edward Cutrell. 2023. Understanding Personalized Accessibility through Teachable AI: Designing and Evaluating Find My Things for People who are Blind or Low Vision. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility* (New York, NY, USA). ACM, 1–12.
- [97] Emily Mower, David J Feil-Seifer, Maja J Mataric, and Shrikanth Narayanan. 2007. Investigating implicit cues for user state estimation in human-robot interaction using physiological measurements. In *RO-MAN 2007-The 16th IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, 1125–1130.
- [98] Amal Nanavati, Vinita Ranganeni, and Maya Cakmak. 2023. Physically assistive robots: A systematic review of mobile and manipulator robots that physically assist people with disabilities. *Annual Review of Control, Robotics, and Autonomous Systems* 7 (2023).
- [99] Isabel Neto, Filipa Correia, Filipa Rocha, Patricia Piedade, Ana Paiva, and Hugo Nicolau. 2023. The robot made us hear each other: Fostering inclusive conversations among mixed-visual ability children. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (Stockholm, Sweden). 13–23.
- [100] Dejing Ni, Aiguo Song, Lei Tian, Xiaonong Xu, and Danfeng Chen. 2015. A walking assistant robotic system for the visually impaired based on computer vision and tactile perception. *International Journal of Social Robotics* 7 (2015), 617–628.
- [101] Dejing Ni, Aiguo Song, Lei Tian, Xiaonong Xu, and Danfeng Chen. 2015. A walking assistant robotic system for the visually impaired based on computer vision and tactile perception. *International Journal of Social Robotics* 7 (2015), 617–628.
- [102] Radosław Niewiadomski, Eleonora Ceccaldi, Gijs Huisman, Gualtiero Volpe, and Maurizio Mancini. 2019. Computational Commensality: from theories to computational models for social food preparation and consumption in HCI. *Frontiers in Robotics and AI* 6 (2019), 119.
- [103] Jagannadh Pariti, Vinita Tibdewal, and Tae Oh. 2020. Intelligent mobility cane—lessons learned from evaluation of obstacle notification system using a haptic approach. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [104] Huang Peng, Guangming Song, Jian You, Ying Zhang, and Jie Lian. 2017. An indoor navigation service robot system based on vibration tactile feedback. *International Journal of Social Robotics* 9 (2017), 331–341.
- [105] Ana Cristina Pires, Filipa Rocha, Antonio José de Barros Neto, Hugo Simão, Hugo Nicolau, and Tiago Guerreiro. 2020. Exploring accessible programming with educators and visually impaired children. In *Proceedings of the Interaction Design and Children Conference*. 148–160.
- [106] David Porfírio, Laura Stegner, Maya Cakmak, Allison Saupé, Aws Albarghouthi, and Bilge Mutlu. 2023. Sketching robot programs on the fly. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (Stockholm, Sweden). 584–593.
- [107] Adil Rahman, Md Aashikur Rahman Azim, and Seongkook Heo. 2023. Take my hand: Automated hand-based spatial guidance for the visually impaired. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 544, 16 pages.
- [108] Roope Raisamo, Katri Salminen, Jussi Rantala, Ahmed Farooq, and Mounia Ziat. 2022. Interpersonal haptic communication: review and directions for the future. *International Journal of Human-Computer Studies* 166 (2022), 102881.
- [109] Vinita Ranganeni, Mike Sinclair, Eyal Ofek, Amos Miller, Jonathan Campbell, Andrey Kolobov, and Edward Cutrell. 2023. Exploring levels of control for a navigation assistant for blind travelers. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (Stockholm, Sweden). 4–12.
- [110] J Redmon. 2016. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- [111] Neil M Richards and William D Smart. 2016. How should the law think about robots? In *Robot law*. Edward Elgar Publishing, 3–22.
- [112] Laurel D Riek. 2012. Wizard of oz studies in hri: a systematic review and new reporting guidelines. *Journal of Human-Robot Interaction* 1, 1 (2012), 119–136.
- [113] Filipa Rocha, Filipa Correia, Isabel Neto, Ana Cristina Pires, João Guerreiro, Tiago Guerreiro, and Hugo Nicolau. 2023. Coding Together: On Co-located and Remote Collaboration between Children with Mixed-Visual Abilities. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 606, 14 pages.
- [114] Filipa Rocha, Ana Cristina Pires, Isabel Neto, Hugo Nicolau, and Tiago Guerreiro. 2021. Assembly at home: Accessible spatial programming for children with visual impairments and their families. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference*. 100–111.
- [115] Andrew Sears and Vicki Hanson. 2011. Representing users in accessibility research. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2235–2238.
- [116] Amanda Sharkey and Noel Sharkey. 2012. Granny and the robots: ethical issues in robot care for the elderly. *Ethics and Information Technology* 14 (2012), 27–40.
- [117] Roy Shilkrot, Jochen Huber, Wong Meng Ee, Pattie Maes, and Suranga Chandima Nanayakkara. 2015. FingerReader: a wearable device to explore printed text on the go. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 2363–2372.
- [118] Roy Shilkrot, Jochen Huber, Jürgen Steimle, Suranga Nanayakkara, and Pattie Maes. 2015. Digital digits: A comprehensive survey of finger augmentation devices. *ACM Computing Surveys (CSUR)* 48, 2 (2015), 1–29.
- [119] Kristen Shinohara and Jacob O Wobbrock. 2011. In the shadow of misperception: assistive technology use and social interactions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 705–714.
- [120] Ben Shneiderman and Catherine Plaisant. 2010. *Designing the User Interface: Strategies for Effective Human-Computer Interaction*. Pearson Education India.
- [121] Bruno Siciliano, Oussama Khatib, and Torsten Kröger. 2008. *Springer handbook of robotics*. Vol. 200. Springer.
- [122] Patrick Slade, Arjun Tambe, and Mykel J Kochenderfer. 2021. Multimodal sensing and intuitive steering assistance improve navigation and mobility for people with impaired vision. *Science robotics* 6, 59 (2021), eabg6594.
- [123] Clay Spinuzzi. 2005. The methodology of participatory design. *Technical Communication* 52, 2 (2005), 163–174.
- [124] Laura Stegner, Emmanuel Senft, and Bilge Mutlu. 2023. Situated participatory design: A method for in situ design of robotic interaction with older adults. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 334, 15 pages.
- [125] Jaimie D Steinmetz, Rupert RA Bourne, Paul Svitil Briant, Seth R Flaxman, Hugh RB Taylor, Jost B Jonas, Amir Aberhe Abdoli, Woldu Aberhe Abbrha, Ahmed Abualhasan, Eman Girum Abu-Gharbieh, et al. 2021. Causes of blindness

- and vision impairment in 2020 and trends over 30 years, and prevalence of avoidable blindness in relation to VISION 2020: the Right to Sight: an analysis for the Global Burden of Disease Study. *The Lancet Global Health* 9, 2 (2021), e144–e160.
- [126] Ryo Suzuki, Adnan Karim, Tian Xia, Hooman Hedayati, and Nicolai Marquardt. 2022. Augmented reality and robotics: A survey and taxonomy for ar-enhanced human-robot interaction and robotic interfaces. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA). ACM, Article 553, 33 pages.
 - [127] Susumu Tachi, Kazuo Tanie, Kiyoshi Komoriya, and Minoru Abe. 1985. Electrocutaneous communication in a guide dog robot (MELDOG). *IEEE Transactions on Biomedical Engineering* 7 (1985), 461–469.
 - [128] Susumu Tachi, Kazuo Tanie, Kiyoshi Komoriya, Yuji Hosoda, and Minoru Abe. 1981. Guide dog robot—its basic plan and some experiments with MELDOG MARK I. *Mechanism and Machine Theory* 16, 1 (1981), 21–29.
 - [129] Nada Terzimehić, Renate Häuslschmid, Heinrich Hussmann, and MC Schraefel. 2019. A review & analysis of mindfulness research in HCI: Framing current lines of research and future opportunities. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland, UK). ACM, Article 457, 13 pages.
 - [130] B Thylefors, AD Negrel, Ramachandra Pararajasegaram, and KY Dadzie. 1995. Global data on blindness. *Bulletin of the World Health Organization* 73, 1 (1995), 115.
 - [131] Cathy Urquhart. 2022. Grounded theory for qualitative research: A practical guide. (2022).
 - [132] John Vines, Rachel Clarke, Peter Wright, John McCarthy, and Patrick Olivier. 2013. Configuring participation: on how we involve people in design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 429–438.
 - [133] Andreas Wachaja, Pratik Agarwal, Mathias Zink, Miguel Reyes Adame, Knut Möller, and Wolfram Burgard. 2015. Navigating blind people with a smart walker. In *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 6014–6019.
 - [134] Andreas Wachaja, Pratik Agarwal, Mathias Zink, Miguel Reyes Adame, Knut Möller, and Wolfram Burgard. 2017. Navigating blind people with walking impairments using a smart walker. *Autonomous Robots* 41 (2017), 555–573.
 - [135] Luyao Wang, Qihe Chen, Yan Zhang, Ziang Li, Tingmin Yan, Fan Wang, Guyue Zhou, and Jiangtao Gong. 2023. Can Quadruped Guide Robots be Used as Guide Dogs?. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 4094–4100.
 - [136] Yang Wang, Huichuan Xia, Yaxing Yao, and Yun Huang. 2016. Flying eyes and hidden controllers: A qualitative study of people's privacy perceptions of civilian drones in the US. *Proceedings on Privacy Enhancing Technologies* (2016).
 - [137] Ron Weber. 2004. Editor's comments: the rhetoric of positivism versus interpretivism: a personal view. *MIS Quarterly* (2004), iii–xii.
 - [138] Lorraine Whitmarsh. 2005. The benefits of guide dog ownership. *Visual Impairment Research* 7, 1 (2005), 27–42.
 - [139] Jacob O Wobbrock, Shaun K Kane, Krzysztof Z Gajos, Susumu Harada, and Jon Froehlich. 2011. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing (TACCESS)* 3, 3 (2011), 1–27.
 - [140] Jacob O Wobbrock and Julie A Kientz. 2016. Research contributions in human-computer interaction. *Interactions* 23, 3 (2016), 38–44.
 - [141] Yan Zhang, Ziang Li, Haole Guo, Luyao Wang, Qihe Chen, Wenjie Jiang, Mingming Fan, Guyue Zhou, and Jiangtao Gong. 2023. "I am the follower, also the boss": Exploring Different Levels of Autonomy and Machine Forms of Guiding Robots for the Visually Impaired. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany). ACM, Article 542, 22 pages.

A APPENDIX A

	Column	Definition
General information of the paper	Name	Name of the paper
	Venue	Venue of the paper
	Year	Publication year of the paper
	Keywords	Index terms or keywords provided by the authors
	RQ/Contributions	Research questions and/or contributions summarized by the authors
	Contribution Type	Contribution type of the paper (based on Wobbrock's framework [140])
	Application Area/Problem	The context and practical problem that the robot is intended to be used
	Target User Group	Target users, highlighting specific groups (e.g., elder blind)
Information about the robot	Functionality	The building blocks of the robot's functionalities
	Input Modality	Methods of user input
	Output Modality	Method of robot output
	Design	Methods and activities used in the design phase
	Implementation (software)	Key technologies in terms of software (e.g., algorithm)
	Implementation (hardware)	Key technologies in terms of hardware (e.g., sensors)
	Embodiment	The robot's physical appearance, size, and weight
	Customizability	Whether and how the robots can be customized or personalized
Evaluation	User study	User study structure
	Metrics/Questions	Quantitative and/or qualitative data collected during evaluation
	Participants (#, age, vision)	Demographics and characteristics of study participants
	Results	Key findings from the evaluation
Other	Limitations/Future Work	Areas for improvement identified by both the authors and our analysis
	Other Comments	Notable observations and additional comments not captured in other categories

Table 3: Definition of extracted information about the surveyed body of papers